

Supplementary Material

Table S1. Parameter design space, engineering constraints, and selected levels

The selected levels were engineering operating points chosen after qualitative screening rather than optimized values or formal physical safety limits. Their intended directional effects describe the controller or API behavior on this platform and in this task; they should not be interpreted as independently identified material properties or fingertip-force measurements.

Parameter	Low-to-high operational effect	Engineering constraint	Selected levels
Slave impedance: K_t	Lower $K_t \rightarrow$ greater translational compliance and lower impact; higher $K_t \rightarrow$ greater positional resistance.	No sustained oscillation during qualitative screening.	50 / 150 / 200 N/m
Slave impedance: K_r	Lower $K_r \rightarrow$ greater rotational compliance; higher $K_r \rightarrow$ greater orientation retention against rotational disturbance.	No sustained oscillation during qualitative screening.	5 / 10 / 13 Nm/rad
Slave impedance: ζ	Increasing $\zeta \rightarrow$ increasing normalized controller damping action for fixed $\mathbf{K}$; it is not a physical critical-damping ratio.	Avoid sustained oscillation and excessive sluggishness.	0.8 / 1.0 / 1.2
Master haptic: K_f	Lower $K_f \rightarrow$ weaker rendered external-force cue; higher $K_f \rightarrow$ stronger cue.	Acceptable Omega.7 response.	0.2 / 0.5 / 0.7
Master haptic: d	Lower $d \rightarrow$ greater sensitivity to small forces; higher $d \rightarrow$ greater suppression of small signals.	Limit master-side haptic jitter and noise.	0.3 / 0.4 / 0.5 N
Gripper: v_g	Lower $v_g \rightarrow$ slower, lower-impact closing; higher $v_g \rightarrow$ faster grasp establishment.	Franka Hand execution range.	0.02 / 0.05 / 0.10 m/s
Gripper: F_g	Lower requested $F_g \rightarrow$ gentler grasping; higher $F_g \rightarrow$ greater requested grasp force.	Franka Hand force setting.	8 / 15 / 20 N

Table S2. Preassigned chronological mode sequences

Operator ID	Group	Mode sequence
P01	1	A $\rightarrow$ D $\rightarrow$ C $\rightarrow$ B $\rightarrow$ E
P01	2	B $\rightarrow$ E $\rightarrow$ D $\rightarrow$ C $\rightarrow$ A
P01	3	C $\rightarrow$ A $\rightarrow$ E $\rightarrow$ D $\rightarrow$ B
P02	4	D $\rightarrow$ B $\rightarrow$ C $\rightarrow$ E $\rightarrow$ A
P02	5	E $\rightarrow$ C $\rightarrow$ A $\rightarrow$ B $\rightarrow$ D
P02	6	A $\rightarrow$ D $\rightarrow$ B $\rightarrow$ C $\rightarrow$ E
P03	7	B $\rightarrow$ E $\rightarrow$ A $\rightarrow$ D $\rightarrow$ C
P03	8	C $\rightarrow$ B $\rightarrow$ D $\rightarrow$ A $\rightarrow$ E
P03	9	D $\rightarrow$ A $\rightarrow$ E $\rightarrow$ C $\rightarrow$ B

The preassigned sequence was applied to all blocks within the corresponding operator group.

Table S3. Class-wise results of the closed-set controlled visual test

Object	Images	Mean confidence	Mean time (ms)
Apple	30	0.771	49.89
Banana	30	0.948	48.02
Bottle	30	0.726	48.57
Paper cup	30	0.820	46.62
Mouse	30	0.914	46.79
Scissors	30	0.938	49.27

All 180 images were classified correctly and mapped to the intended strategy under the fixed viewpoint, background, and lighting conditions.

Table S4. Trial-level performance data

Each row corresponds to a single trial. T denotes completion time (s), L denotes master-side trajectory length (m), and the outcome is coded as S (success) or F (failure). Rows are grouped by block and listed in analytical mode order (A–E), not chronological order; the preassigned chronological mode sequence for each operator group is reported in Table S2.

Block ID	Operator ID	Group	Object	Mode	T (s)	L (m)	Outcome
MB01	P01	1	apple	A	19.7647	0.6263	S
MB01	P01	1	apple	B	20.3613	0.5712	F
MB01	P01	1	apple	C	18.2302	0.7079	S
MB01	P01	1	apple	D	20.1232	0.7471	S
MB01	P01	1	apple	E	18.8263	0.7321	S
MB02	P01	1	paper cup	A	20.2387	0.6952	S
MB02	P01	1	paper cup	B	18.8340	0.6637	S
MB02	P01	1	paper cup	C	18.0432	0.5838	S
MB02	P01	1	paper cup	D	19.1196	0.6777	F
MB02	P01	1	paper cup	E	19.9499	0.6712	S
MB03	P01	1	mouse	A	21.9673	0.6332	F
MB03	P01	1	mouse	B	17.3734	0.6484	S
MB03	P01	1	mouse	C	16.4537	0.5699	S
MB03	P01	1	mouse	D	20.7266	0.7682	S
MB03	P01	1	mouse	E	22.2490	0.6481	S
MB04	P01	2	banana	A	19.0863	0.7552	S
MB04	P01	2	banana	B	20.4131	0.8350	S
MB04	P01	2	banana	C	19.5678	0.6649	S
MB04	P01	2	banana	D	20.0671	0.6714	S
MB04	P01	2	banana	E	20.3914	0.8316	S
MB05	P01	2	paper cup	A	25.9611	0.9493	S
MB05	P01	2	paper cup	B	20.1454	0.7550	S
MB05	P01	2	paper cup	C	17.8797	0.6381	S
MB05	P01	2	paper cup	D	20.7920	0.8015	S
MB05	P01	2	paper cup	E	21.0563	0.7940	S
MB06	P01	2	mouse	A	22.1784	0.7902	S
MB06	P01	2	mouse	B	19.3855	0.6696	F
MB06	P01	2	mouse	C	19.6104	0.6812	S
MB06	P01	2	mouse	D	20.6688	0.7218	S
MB06	P01	2	mouse	E	22.0362	0.7554	S
MB07	P01	3	banana	A	21.3677	0.8979	S
MB07	P01	3	banana	B	20.7620	0.7810	S
MB07	P01	3	banana	C	20.6135	0.6701	S
MB07	P01	3	banana	D	21.3607	0.7123	S
MB07	P01	3	banana	E	18.8917	0.7928	S
MB08	P01	3	bottle	A	21.1942	0.7547	S
MB08	P01	3	bottle	B	20.6644	0.6744	S
MB08	P01	3	bottle	C	19.5792	0.6695	S
MB08	P01	3	bottle	D	20.9106	0.7339	F
MB08	P01	3	bottle	E	18.3977	0.6920	S
MB09	P01	3	scissors	A	20.6590	0.5952	S
MB09	P01	3	scissors	B	20.8892	0.9875	F
MB09	P01	3	scissors	C	20.4931	0.7578	S

Table S4 (continued)

Block ID	Operator ID	Group	Object	Mode	T (s)	L (m)	Outcome
MB09	P01	3	scissors	D	20.8983	0.5998	S
MB09	P01	3	scissors	E	23.6364	0.7184	F
MB10	P02	4	apple	A	20.9540	0.7275	S
MB10	P02	4	apple	B	22.4177	0.7386	S
MB10	P02	4	apple	C	19.9973	0.8459	S
MB10	P02	4	apple	D	20.3035	0.6780	S
MB10	P02	4	apple	E	22.2594	0.8061	S
MB11	P02	4	bottle	A	21.6166	0.6533	S
MB11	P02	4	bottle	B	22.7903	0.8687	S
MB11	P02	4	bottle	C	18.7488	0.6005	S
MB11	P02	4	bottle	D	20.6251	0.7235	S
MB11	P02	4	bottle	E	18.9052	0.6659	S
MB12	P02	4	scissors	A	22.1109	0.8229	S
MB12	P02	4	scissors	B	20.1043	0.7655	F
MB12	P02	4	scissors	C	17.1904	0.6565	S
MB12	P02	4	scissors	D	24.4447	0.6999	S
MB12	P02	4	scissors	E	20.7281	0.6800	S
MB13	P02	5	apple	A	20.6792	0.7170	S
MB13	P02	5	apple	B	20.8904	0.8292	S
MB13	P02	5	apple	C	18.3496	0.8878	S
MB13	P02	5	apple	D	21.1041	0.7061	S
MB13	P02	5	apple	E	20.5426	0.8040	S
MB14	P02	5	bottle	A	21.1828	0.7639	F
MB14	P02	5	bottle	B	21.4721	0.6454	S
MB14	P02	5	bottle	C	21.3193	0.7792	S
MB14	P02	5	bottle	D	19.9772	0.6640	S
MB14	P02	5	bottle	E	23.9520	0.6377	F
MB15	P02	5	scissors	A	21.1103	0.6776	F
MB15	P02	5	scissors	B	19.7492	0.7763	F
MB15	P02	5	scissors	C	17.5832	0.7542	S
MB15	P02	5	scissors	D	20.9640	0.6468	S
MB15	P02	5	scissors	E	22.8528	0.5349	S
MB16	P02	6	banana	A	20.8514	0.8434	S
MB16	P02	6	banana	B	20.2425	1.0097	S
MB16	P02	6	banana	C	19.1901	0.9025	S
MB16	P02	6	banana	D	22.9636	0.7686	S
MB16	P02	6	banana	E	22.5321	0.9161	S
MB17	P02	6	paper cup	A	23.1561	0.7838	F
MB17	P02	6	paper cup	B	21.0744	0.7871	S
MB17	P02	6	paper cup	C	19.8447	0.5853	S
MB17	P02	6	paper cup	D	22.3201	0.8671	S
MB17	P02	6	paper cup	E	20.6399	0.7132	S
MB18	P02	6	mouse	A	22.8171	0.6597	S
MB18	P02	6	mouse	B	19.1697	0.8023	S
MB18	P02	6	mouse	C	19.6294	0.8223	S
MB18	P02	6	mouse	D	20.7460	0.8665	S
MB18	P02	6	mouse	E	22.5251	0.6652	S
MB19	P03	7	apple	A	20.4341	0.7572	S
MB19	P03	7	apple	B	21.9394	0.8520	S
MB19	P03	7	apple	C	20.4061	0.6901	S
MB19	P03	7	apple	D	21.7007	0.7223	S
MB19	P03	7	apple	E	22.1380	0.7477	S
MB20	P03	7	paper cup	A	20.9114	0.9774	S
MB20	P03	7	paper cup	B	21.4933	0.9173	F
MB20	P03	7	paper cup	C	18.5434	0.7043	S
MB20	P03	7	paper cup	D	20.8499	0.7190	S
MB20	P03	7	paper cup	E	20.6803	0.7488	F
MB21	P03	7	mouse	A	22.0561	0.6905	S
MB21	P03	7	mouse	B	24.5042	0.8406	S
MB21	P03	7	mouse	C	22.2288	0.7984	F
MB21	P03	7	mouse	D	20.3404	0.8234	S
MB21	P03	7	mouse	E	21.4320	0.7381	S
MB22	P03	8	banana	A	20.5840	0.8097	S
MB22	P03	8	banana	B	21.7273	0.9022	S
MB22	P03	8	banana	C	19.6026	0.7299	S

Table S4 (continued)

Block ID	Operator ID	Group	Object	Mode	T (s)	L (m)	Outcome
MB22	P03	8	banana	D	20.2005	0.7632	S
MB22	P03	8	banana	E	21.2047	0.8606	S
MB23	P03	8	paper cup	A	21.5467	0.8037	S
MB23	P03	8	paper cup	B	19.7902	0.7035	S
MB23	P03	8	paper cup	C	19.1190	0.6475	S
MB23	P03	8	paper cup	D	18.8053	0.6170	S
MB23	P03	8	paper cup	E	23.5113	0.6767	S
MB24	P03	8	mouse	A	19.7035	0.8352	S
MB24	P03	8	mouse	B	24.2040	1.0190	S
MB24	P03	8	mouse	C	20.8218	0.7430	S
MB24	P03	8	mouse	D	21.2149	0.6921	S
MB24	P03	8	mouse	E	19.9535	0.7165	S
MB25	P03	9	banana	A	18.4814	0.7055	S
MB25	P03	9	banana	B	21.0957	0.9070	S
MB25	P03	9	banana	C	18.4779	0.8467	S
MB25	P03	9	banana	D	20.7293	0.6453	S
MB25	P03	9	banana	E	20.1558	0.8482	S
MB26	P03	9	bottle	A	24.4515	0.9096	F
MB26	P03	9	bottle	B	23.6510	0.7651	S
MB26	P03	9	bottle	C	18.8794	0.6972	S
MB26	P03	9	bottle	D	21.6030	0.8028	S
MB26	P03	9	bottle	E	19.9488	0.8517	S
MB27	P03	9	scissors	A	23.2623	0.7705	S
MB27	P03	9	scissors	B	22.0770	0.8524	S
MB27	P03	9	scissors	C	20.0995	0.6642	S
MB27	P03	9	scissors	D	21.0219	0.9843	F
MB27	P03	9	scissors	E	19.5680	0.6959	S

Table S5. Mean completion time by object and experimental mode

Object	A	B	C	D	E
Apple	20.46	21.40	19.25	20.81	20.94
Banana	20.07	20.85	19.49	21.06	20.64
Paper cup	22.36	20.27	18.69	20.38	21.17
Bottle	22.11	22.14	19.63	20.78	20.30
Mouse	21.74	20.93	19.75	20.74	21.64
Scissors	21.79	20.70	18.84	21.83	21.70

Values are mean completion times in seconds; bold indicates the lowest mean in each row.

Table S6. Evidence-boundary overview

Claim	Direct evidence in the present study	Supported interpretation
Complete multi-channel configuration was associated with a lower mean task completion time than impedance-only reconfiguration.	The paired C–E mean difference was 1.79 s (matched-block bootstrap 95% CI, 1.10–2.51 s); 22 of 27 matched blocks favored mode C.	Supported as a system-level, within-sample result under the tested conditions.
Mode C has a higher success rate.	Observed success was 26/27 in C and 24/27 in E. The paired outcomes were 23 both-success blocks, three C-success/E-failure blocks, and one C-failure/E-success block; the exact two-sided McNemar test gave $p = 0.625$.	Descriptive only; the paired success comparison did not establish a mode effect.
Mode C reduces workload.	Raw NASA-TLX was lower in C in all nine operator $\times$ strategy units.	Descriptive within the present sample; no population-level workload inference.
The time benefit results from a shorter path or fewer pauses.	The trajectory-length bootstrap interval crossed zero. Pause-count medians were equal, and no threshold-sensitivity analysis was available.	Not established; these measures are exploratory and do not identify a mechanism.
The master-side haptic and gripper-execution channels each contribute independently, or their combination is synergistic.	Modes C and E changed the master-side haptic and gripper-execution parameter channels together; no factorial channel-ablation design was included.	Not tested.
Configuration was completed before contact in every trial, or the implementation is hard real-time.	Formal trials lacked synchronized trigger–contact timestamps, and hard real-time performance was not evaluated.	Not verified.

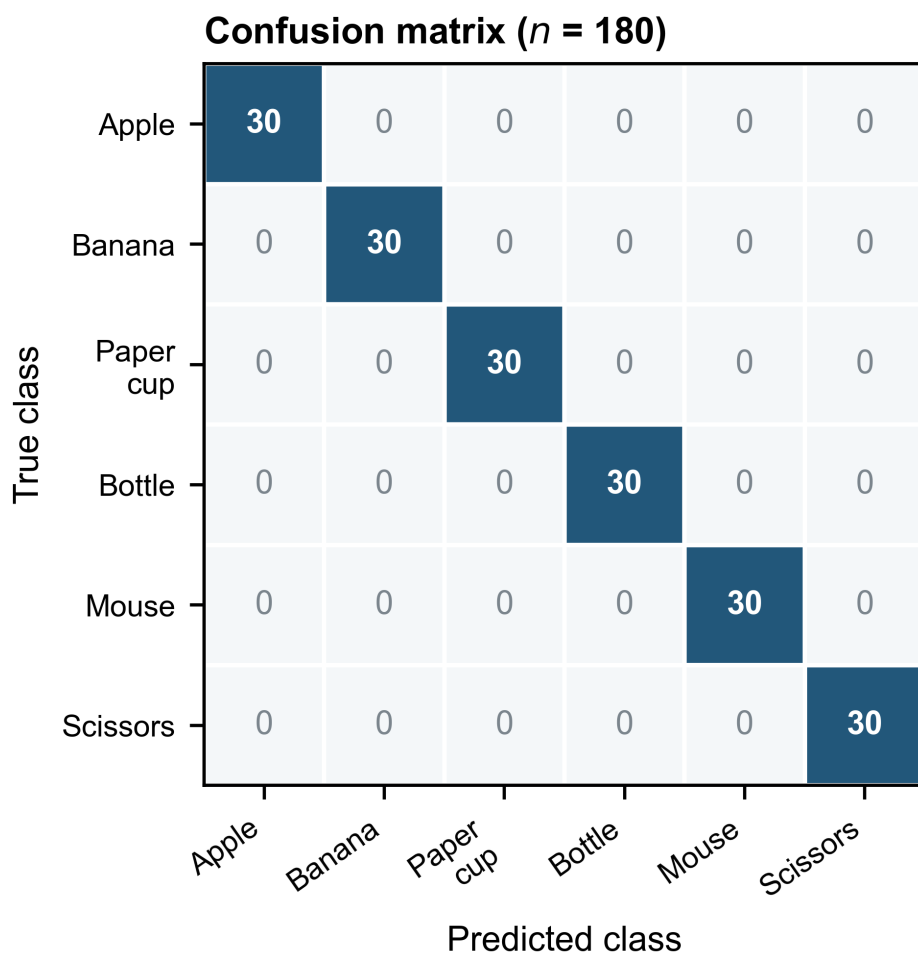


Figure S1. Confusion matrix for the separate closed-set controlled visual test (180 images; 30 images per class). All observations lie on the diagonal under the fixed viewpoint, background, and lighting conditions.